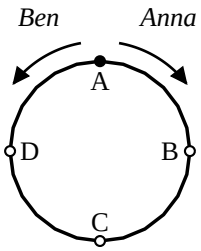
Math Kangaroo Practice for Koray

Grade 3-4 · 20 questions · made from Koray's practice · seed 68320

Part A: Each question is worth 3 points

1. Anna and Ben draw a big circle in the sand and write the letters A, B, C and D on it. They start at the same time from point A. Anna runs clockwise and Ben runs anticlockwise around the circle (see picture). They meet for the first time at point D, then at C, at B and then again at point A. How many times has Anna run around the circle by then?



A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 8 times

2. Which number has the same sum of digits as 912?

A. 652 ***B.*** 1057 ***C.*** 1064 ***D.*** 1155 ***E.*** 1156

3. A theatre has 33 seats, numbered from 1 to 33. Tom writes all the seat numbers on cards, one digit on each card.
How many cards does Tom need?

A. 33 ***B.*** 57 ***C.*** 60 ***D.*** 64 ***E.*** 66

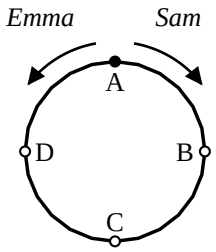
4. The odometer of a car shows 62958 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 2 ***B.*** 11 ***C.*** 12 ***D.*** 15 ***E.*** 21

5. Mia writes all the numbers from 96 to 106.
How many digits does Mia write?

A. 20 ***B.*** 29 ***C.*** 30 ***D.*** 33 ***E.*** 36

6. Sam and Emma draw a big circle in the sand and write the letters A, B, C and D on it. They start at the same time from point A. Sam runs clockwise and Emma runs anticlockwise around the circle (see picture). They meet for the first time at point D, then at C, at B and then again at point A. How many times has Sam run around the circle by then?

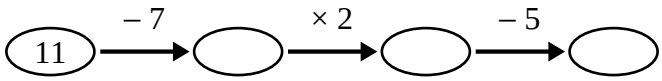


A. 3 times ***B.*** 4 times ***C.*** 5 times ***D.*** 6 times ***E.*** 8 times

7. Which number has the same sum of digits as 872?

A. 646 ***B.*** 718 ***C.*** 772 ***D.*** 1087 ***E.*** 1097

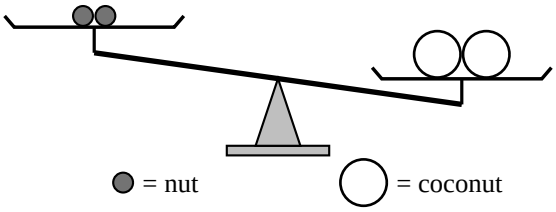
8. Follow the arrows and do each calculation in turn.



Which number comes out at the end?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 11 ***E.*** 14

9. On one plate of a balance there are 2 nuts and on the other there are 2 coconuts. When we put one more coconut, exactly like the others, on the plate with the nuts, the balance is level.



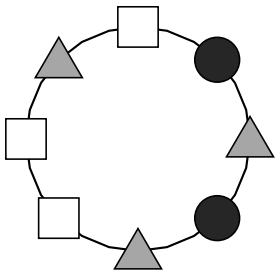
One coconut weighs the same as how many nuts?

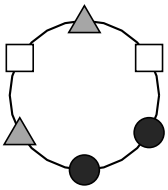
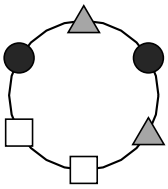
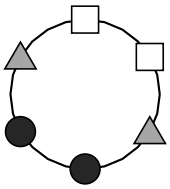
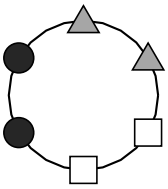
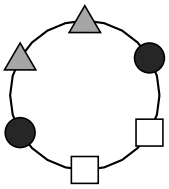
A. 5 nuts ***B.*** 4 nuts ***C.*** 3 nuts ***D.*** 2 nuts ***E.*** 1 nut

10. Omar, who is older than Leo by 1 year and 1 day, was born on February 28, 2003. When was Leo born?

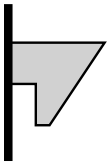
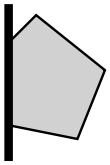
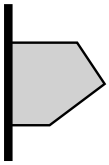
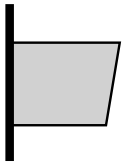
A. March 1, 2005 ***B.*** February 27, 2004 ***C.*** February 29, 2004
D. February 28, 2004 ***E.*** March 1, 2002

11. The picture shows a bracelet with circles, squares and triangles. Leo removes two neighbouring stones: one square and one triangle. Which bracelet can Leo get?

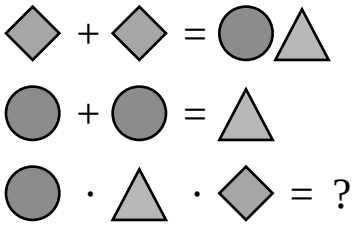


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

12. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Nina draws the shadow at five different moments.
Which picture **cannot** be right unless the flag is torn?

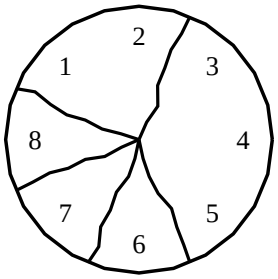
A.  ***B.***  ***C.*** ***D.***  ***E.*** 

13. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Anna lays out the cards so that the calculations are correct. What number does Anna get for the last calculation?



A. 0 ***B.*** 9 ***C.*** 12 ***D.*** 13 ***E.*** 24

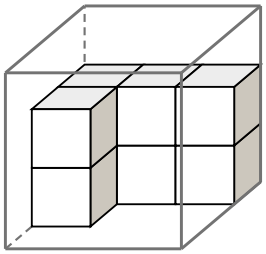
14. A wheel of fortune with the numbers 1 to 8 around its edge is cracked into 5 pieces as shown.



Which piece has the largest sum of numbers?

A. the piece with the 6 ***B.*** the piece with the 5 ***C.*** the piece with the 7
D. the piece with the 8 ***E.*** the piece with the 2

15. Mia has a glass aquarium shaped like a cube, 3 dm long, 3 dm wide and 3 dm tall. Mia has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



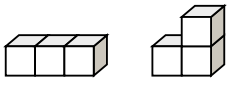
At most how many more such cubes can Mia put into the aquarium?

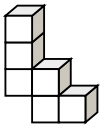
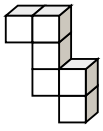
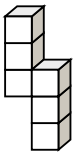
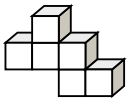
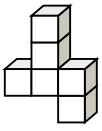
A. 18 ***B.*** 19 ***C.*** 20 ***D.*** 22 ***E.*** 27

16. Alex has the cards 3, 5 and 8. Alex must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Alex get?

A. 35 ***B.*** 38 ***C.*** 41 ***D.*** 43 ***E.*** 45

17. Kanga wants to build a flat figure out of these 2 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures **can** he make?

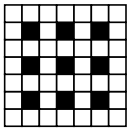


A. 
B. 
C. 
D. 
E. 

18. These diagrams follow a pattern of black and white cells. If the pattern continues, how many white cells will the next diagram have?

8 white cells

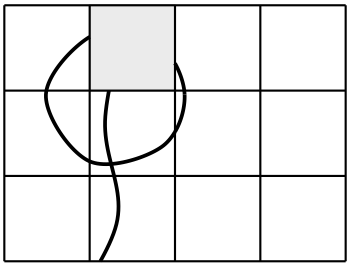
21 white cells



40 white cells

A. 65 ***B.*** 69 ***C.*** 70 ***D.*** 72 ***E.*** 81

19. Mia drew some lines on squared paper. Then she cut out one square of her picture.



Which square was cut out of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

20. Jumpy the kangaroo's longest jump in training was 30 dm 40 cm and 70 mm long. At the competition the jump was 123 cm longer.
How long was the jump at the competition?

A. 3 m 47 cm ***B.*** 4 m 70 cm ***C.*** 5 m 33 cm ***D.*** 5 m 70 cm ***E.*** 15 m 77 cm

Answer Key

1.	**C**	**2.**	**D**	**3.**	**B**	**4.**	**C**	**5.**	**B**	**6.**	**A**
7.	**E**	**8.**	**B**	**9.**	**D**	**10.**	**C**	**11.**	**B**	**12.**	**A**
13.	**C**	**14.**	**B**	**15.**	**B**	**16.**	**D**	**17.**	**A**	**18.**	**A**
19.	**A**	**20.**	**B**								

Solutions

1. C (3 times) — Between two meetings they run exactly once around the circle together. Ben only gets to the next letter (1 of 4 parts), so Anna runs the other 3 parts. 4 meetings × 3 parts = 12 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

2. D (1155) — 9 + 1 + 2 = 12 and 1 + 1 + 5 + 5 = 12. The others add up to 11 or 13. [set-03/by_level/Grade_03-04/2007 Q4 · which_same_sum]

3. B (57) — 9 × 1 + 24 × 2 = 57. [set-03/by_level/Grade_03-04/2003 Q16 · digits_for_pages]

4. C (12) — Counting up, every number before 62970 repeats a digit. 62970 has all different digits: 62970 – 62958 = 12. [set-03/by_level/Grade_03-04/2002 Q24 · all_different]

5. B (29) — There are 11 numbers: 4 × 2 + 7 × 3 = 29 digits. [set-03/by_level/Grade_03-04/2007 Q23 · across_hundred]

6. A (3 times) — Between two meetings they run exactly once around the circle together. Emma only gets to the next letter (1 of 4 parts), so Sam runs the other 3 parts. 4 meetings × 3 parts = 12 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

7. E (1097) — 8 + 7 + 2 = 17 and 1 + 0 + 9 + 7 = 17. The others add up to 16 or 18. [set-03/by_level/Grade_03-04/2007 Q4 · which_same_sum]

8. B (3) — 11 – 7 = 4, 4 × 2 = 8, 8 – 5 = 3. [set-03/by_level/Grade_03-04/2007 Q5 · forwards]

9. D (2 nuts) — Take 1 coconut off both plates: 2 nuts balance 2 – 1 = 1 coconut. So one coconut = 2 nuts. [set-03/by_level/Grade_03-04/2002 Q9 · one_more_balances]

10. C (February 29, 2004) — Older means born earlier, so Leo was born 1 year and 1 day **after** February 28, 2003: 1 year later is February 28, 2004, then 1 more day: February 29, 2004. [set-03/by_level/Grade_03-04/2007 Q12 · younger_birthday]

11. B — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group square, triangle leaves the bracelet in B, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

12. A — The side sewn to the pole always shows at its full length. In picture A part of the pole is bare, so a piece of the flag is missing. [set-03/by_level/Grade_03-04/2002 Q15 · torn_flag]

13. C (12) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: circle = 1, diamond = 6, triangle = 2. So 1 · 2 · 6 = 12. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

14. B (the piece with the 5) — The sums are: 3+4+5 = 12, 6 = 6, 7 = 7, 8 = 8, 1+2 = 3. [set-03/by_level/Grade_03-04/2002 Q11 · biggest_piece]

15. B (19) — The aquarium holds 3 × 3 × 3 = 27 cubes. There are already 8 inside (count column by column, also the hidden ones). 27 – 8 = 19. [set-03/by_level/Grade_03-04/2007 Q14 · how_many_more]

16. D (43) — 3 + 5 × 8 = 43. None of the other numbers can be made. [set-03/by_level/Grade_03-04/2002 Q21 · which_can]

17. A — Look for one part inside the figure; what is left must be the other part. That works only for figure A. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · two_parts]

18. A (65) — The sides grow by 2 and the black cells are 1 × 1, 2 × 2, 3 × 3, ... The next diagram is 9 × 9 = 81 cells with 4 × 4 = 16 black: 81 – 16 = 65. [set-03/by_level/Grade_03-04/2007 Q21 · next_white]

19. A — Look where the lines of the picture reach the grey gap: top: 1 line, right: 1 line, bottom: 1 line, left: 1 line. Only square A has its lines in exactly those places without being turned. The others are from another place, upside down. [set-03/by_level/Grade_03-04/2002 Q1 · missing_square]

20. B (4 m 70 cm) — 30 dm = 3 m and 70 mm = 7 cm, so the training jump was 3 m 47 cm. Adding 123 cm gives 4 m 70 cm. [set-03/by_level/Grade_03-04/2003 Q10 · jump]